# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE NAME: CSA04 COURSE CODE: Operating Systems**

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**COURSE OUTCOME COVERED**

***CO1:*** *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.*

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**QUESTIONS: 1** **Total Marks:100**

Annexure A

**Technical Presentation**

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***Code of Conduct:***

*I* P.Saketh Kumar Reddy (192525240) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

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Annexure A

## Technical Presentation [Any one]

**Question 1**
Suppose an Operating System is used in a banking system handling thousands of transactions simultaneously.
Prepare a technical presentation explaining:
Types of OS suitable for this scenario
OS structure used
Role of system calls in transaction processing **(BL3)**

**Question 2**
Consider a multi-user Linux-based server environment supporting file sharing and device access.
Prepare a technical presentation covering:
OS functions involved in resource management
File and device system calls
Interaction between user processes and kernel **(BL3)**

**Question 3**
Assume a real-time embedded system used in traffic signal control.
Prepare a technical presentation explaining:
Type of OS used and justification
OS structure and components
System calls used for process control and timing **(BL3)**

**Question 4**
A company uses a distributed computing system across multiple locations.
Prepare a technical presentation discussing:
Type of OS and its characteristics
Architecture of distributed OS
System calls used for communication and coordination **(BL3)**

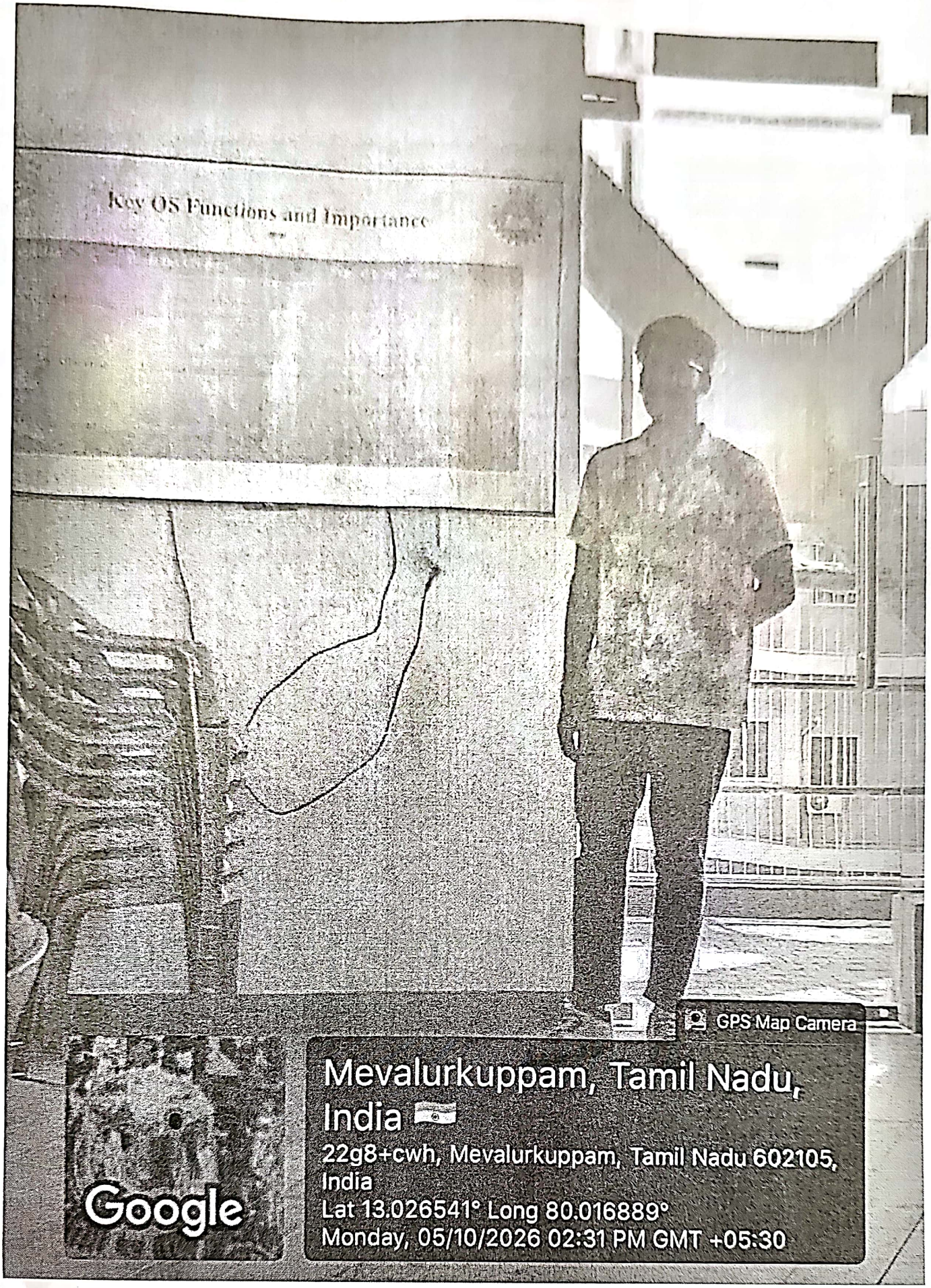
**Question 5**
Consider an OS managing process scheduling, memory allocation, and I/O operations in a modern computer.
Prepare a technical presentation including:
Key OS functions and their importance
Interaction between OS components
Examples of system calls for each function **(BL3)**

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## RUBRICS FOR CALCULATION

| Technical Presentation | | | | | |
|---|---|---|---|---|---|
| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
| Technical Content Accuracy | 30 | Highly accurate, in-depth, and insightful technical explanation | Technically correct content with adequate depth and clarity | Basic concepts presented with limited accuracy and depth | Content contains major technical errors; concepts poorly understood |
| Problem Identification | 25 | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach | Problem identified but analysis or methodology is weak | Problem statement unclear or incorrect; no logical approach |
| Use of Tools | 20 | Advanced tools, well-analyzed data, and highly effective visuals | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation | Minimal or incorrect use of tools and data; poor visuals |
| Communication | 15 | Highly engaging, confident, and audience-focused delivery | Clear, organized, and professional communication | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides |
| Professionalism | 10 | Exemplary professionalism, leadership, and ethical awareness | Professional conduct with effective team collaboration | Basic professionalism; uneven team participation | Lacks professionalism; minimal contribution or ethical awareness |

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